**SIMATS ENGINEERING**

**ASSESSMENT TOOL III— Interactive Dashboard Project**

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|---|---|---|
| **Course: Data Handling and Visualization** | **Code: DSA06** | **CO: CO5** |
| **Duration: 180 min** | **Total Marks: 100** | **Reg No : 192524108**<br>**Name: S. LAKSHITA** |

## COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods, applied through the design of interactive dashboards. (BL4)

**Activity Tool : Interactive Dashboard Project**

**Weightage: 20%**

**Code of Conduct:**

I S.LAKSHITA(192524108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Assessment Criteria**

| Criteria | Dashboard Design and Understanding 25% | Visual Analysis and Interpretation 25% | Application of Dashboard Design Principles 20% | Organization and Presentation 20% | Accuracy 10% | Clarity 5% | Total |
|---|---|---|---|---|---|---|---|
| Q1 | | | | | | | |
| Q2 | | | | | | | |
| Q3 | | | | | | | |
| Q4 | | | | | | | |
| Q5 | | | | | | | |

**Interactive Dashboard Project — Assessment Tool**

**Tableau Practical Questions with Datasets**

**Instructions:** Answer all five questions. Use Tableau to construct the required interactive dashboards, analyze the patterns, compare design choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

### Q1. Building a Sales Performance Dashboard

A retail company records regional sales performance across product categories. Using Tableau, connect to the dataset below and build an interactive dashboard containing: a bar chart of Sales by Region, a line chart of Sales trend by Month, and a map view of Sales by Region. Add a Region filter that acts as a global filter across all views. Analyze which region and category combinations drive the highest sales, and explain how the dashboard layout supports quick comparison across regions.

**Data:**

| S.No | Region | Category | Sales |
|---|---|---|---|
| 1 | North | Furniture | 32000 |
| 2 | South | Furniture | 28500 |
| 3 | East | Furniture | 41000 |
| 4 | West | Furniture | 25500 |
| 5 | North | Technology | 51000 |
| 6 | South | Technology | 47500 |
| 7 | East | Technology | 62000 |
| 8 | West | Technology | 39500 |
| 9 | North | Office Supplies | 21000 |
| 10 | South | Office Supplies | 19500 |
| 11 | East | Office Supplies | 27000 |
| 12 | West | Office Supplies | 17500 |

*Expected Tableau tasks: data connection and preparation, bar chart and line chart worksheets, map visualization, global filter action, dashboard layout and formatting, interpretation of regional and category patterns.*

### Q2. KPI Dashboard with Parameters

An HR analytics team wants to monitor employee performance across departments. Using the dataset below, build a Tableau dashboard containing KPI summary cards (Average Rating, Total Employees, Average Attendance) and a bar chart of Average

Rating by Department. Add a parameter control that lets the user switch the bar chart measure between Average Rating and Average Attendance. Explain how parameters improve the dashboard's flexibility compared to using separate static charts.

**Data:**

| S.No | Department | Rating | Attendance |
|---|---|---|---|
| 1 | Sales | 7.8 | 92 |
| 2 | Sales | 8.1 | 95 |
| 3 | Marketing | 7.5 | 88 |
| 4 | Marketing | 7.9 | 90 |
| 5 | Engineering | 8.6 | 96 |
| 6 | Engineering | 8.9 | 97 |
| 7 | Support | 7.2 | 85 |
| 8 | Support | 7.4 | 87 |
| 9 | HR | 8 | 93 |
| 10 | HR | 8.3 | 94 |

*Expected Tableau tasks: calculated fields for KPI cards, parameter creation, calculated field driven by parameter, bar chart worksheet, KPI card formatting, dashboard assembly, interpretation of flexibility gained from parameters.*

**Q3. Drill-Down Dashboard with Dashboard Actions**

A college wants to analyze student enrollment across Program and Specialization levels. Using the hierarchical dataset below, build a Tableau dashboard with a summary chart at the Program level and a detail chart at the Specialization level. Configure a filter action so that clicking a bar in the Program chart updates the Specialization chart to show only the relevant rows. Analyze how the drill-down interaction improves exploration compared to showing all specializations at once, and note any limitations of this approach.

**Data:**

| S.No | Program | Specialization | Enrollment |
|---|---|---|---|
| 1 | CSE | AI/ML | 65 |

| | | | |
|---|---|---|---|
| 2 | CSE | Cyber Security | 40 |
| 3 | CSE | Core CSE | 90 |
| 4 | ECE | VLSI | 35 |
| 5 | ECE | Communications | 50 |
| 6 | MECH | Robotics | 30 |
| 7 | MECH | Thermal | 45 |
| 8 | MECH | Design | 28 |

*Expected Tableau tasks: hierarchy or two related worksheets, filter action configuration between sheets, dashboard interactivity testing, interpretation of drill-down behavior and its limitations.*

**Q4. Multivariate Analysis Dashboard**

A marketing study records advertising expenditure, website visits, and sales for 15 observations. Build a Tableau dashboard with a scatterplot of Advertising versus Sales, using color or size encoding to represent Website Visits, along with a trend line. Add a second view (box plot or histogram) summarizing the distribution of Sales. Analyze the strength and direction of the relationship shown in the scatterplot and discuss whether the additional Website Visits encoding adds insight or clutters the dashboard.

**Data:**

| S.No | Advertising | Visits | Sales |
|---|---|---|---|
| 1 | 10 | 120 | 48 |
| 2 | 12 | 135 | 50 |
| 3 | 14 | 150 | 53 |
| 4 | 16 | 165 | 55 |
| 5 | 18 | 180 | 58 |
| 6 | 20 | 195 | 61 |
| 7 | 22 | 210 | 63 |
| 8 | 24 | 225 | 66 |
| 9 | 26 | 245 | 69 |

| | | | |
|---|---|---|---|
| 10 | 28 | 260 | 71 |
| 11 | 30 | 275 | 74 |
| 12 | 32 | 290 | 77 |
| 13 | 34 | 305 | 80 |
| 14 | 36 | 320 | 83 |
| 15 | 38 | 340 | 86 |

*Expected Tableau tasks: scatterplot with trend line, color/size encoding for a third variable, secondary distribution view, dashboard layout with legends, relationship analysis and critique of visual clutter.*

**Q5. Misleading Dashboard and Redesign**

A college reports the following pass percentages for seven departments. Build two Tableau dashboards: (1) a misleading version using a truncated y-axis, an inappropriate color scheme, and exaggerated visual emphasis on one department, and (2) a redesigned version using an appropriate baseline, ordered categories, consistent color encoding, and clear labels. Compare the two dashboards and explain how the first design can mislead viewers, and why the redesigned dashboard communicates the data more honestly.

**Data:**

| S.No | Department | Pass_Percentage |
|---|---|---|
| 1 | CSE | 91 |
| 2 | ECE | 93 |
| 3 | EEE | 92 |
| 4 | MECH | 89 |
| 5 | CIVIL | 90 |
| 6 | AIDS | 95 |
| 7 | IT | 94 |

## Q1. Building a Sales Performance Dashboard

**Aim**

To create an interactive Tableau dashboard to analyze regional and category-wise sales performance.

**Algorithm**

1. Connect the sales dataset to Tableau.
2. Create a bar chart for Sales by Region.
3. Create a line chart for Sales trend by Month.
4. Create a map showing Sales by Region.
5. Add Region as a global filter.
6. Combine all views into a dashboard.
7. Analyze the regional and category-wise sales patterns.

**Tableau Procedure**

- Drag **Region** to Columns and **Sales** to Rows to create the bar chart.
- Drag **Region** to the map view and use **Sales** for size/color.
- Create the monthly sales trend using **Month** and **Sales**.
- Add **Region** as a filter.
- Select **Apply to Worksheets → All Using This Data Source**.
- Arrange all charts in a single dashboard.

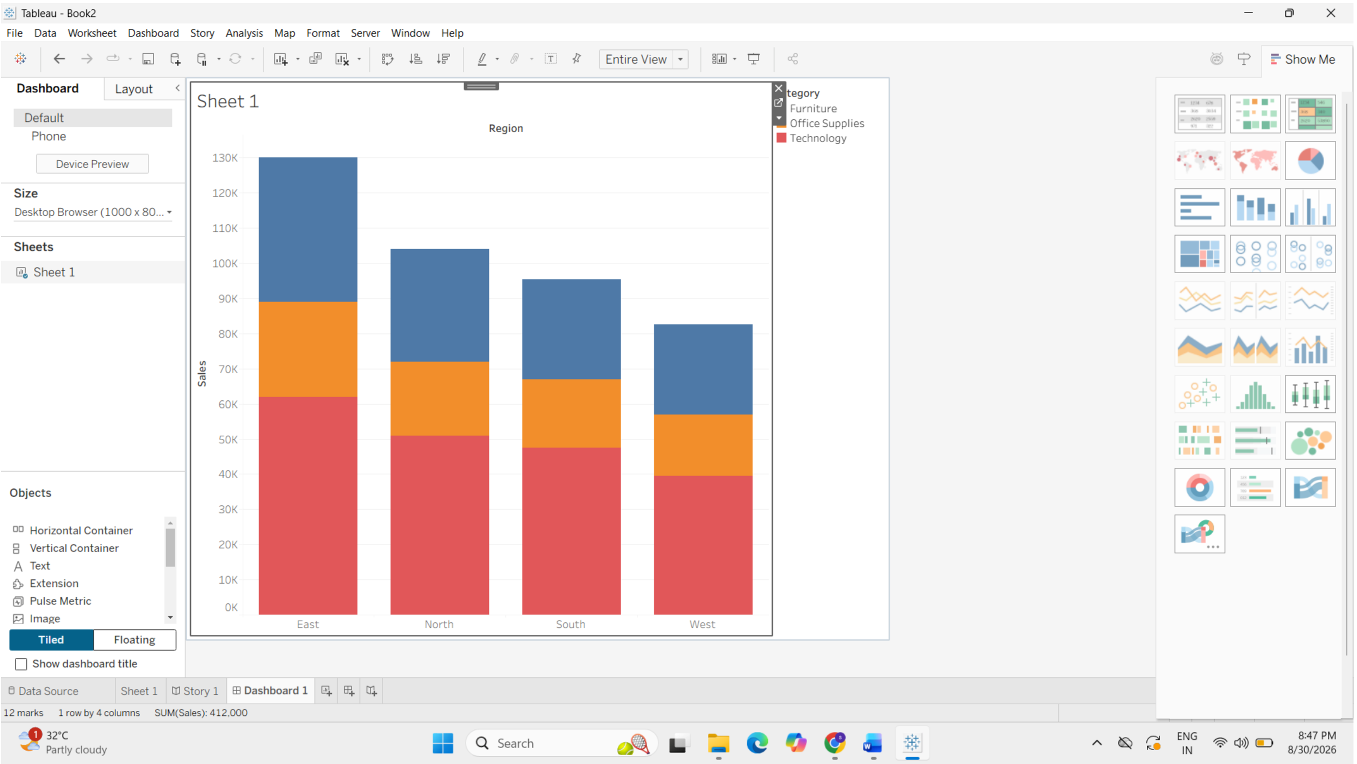
**Analysis**

East region has the highest total sales of **130,000**, while West has the lowest total sales of **82,500**. Technology is the highest-selling category in all regions. The highest region-category combination is **East–Technology with sales of 62,000**. The dashboard layout supports quick comparison by displaying regional sales through bars and maps and allowing users to filter regions interactively.

**Result**

Thus, an interactive Sales Performance Dashboard was created successfully in Tableau.

**Note:** The given dataset does not contain a Month field, so an actual monthly sales trend cannot be calculated from the supplied data.




## Q2. KPI Dashboard with Parameters

### Aim

To create an interactive HR KPI dashboard using Tableau parameters to compare employee performance measures.

### Algorithm

1. Connect the HR dataset to Tableau.
2. Calculate Average Rating.
3. Calculate Total Employees.
4. Calculate Average Attendance.
5. Create KPI cards.
6. Create a parameter to select Rating or Attendance.
7. Create a bar chart based on the selected parameter.
8. Combine the charts into a dashboard.

### Calculated Fields

### Average Rating

AVG([Rating])

**Total Employees**

COUNT([Department])

**Average Attendance**

AVG([Attendance])

**Parameter Calculation**

```
IF [Select Measure] = "Average Rating"
THEN AVG([Rating])
ELSE AVG([Attendance])
END
```

**KPI Results**

- **Average Rating = 7.97**
- **Total Employees = 10**
- **Average Attendance = 91.7%**

**Department Analysis**

Sales: Rating = **7.95**, Attendance = **93.5%**

Marketing: Rating = **7.70**, Attendance = **89%**

Engineering: Rating = **8.75**, Attendance = **96.5%**

Support: Rating = **7.30**, Attendance = **86%**

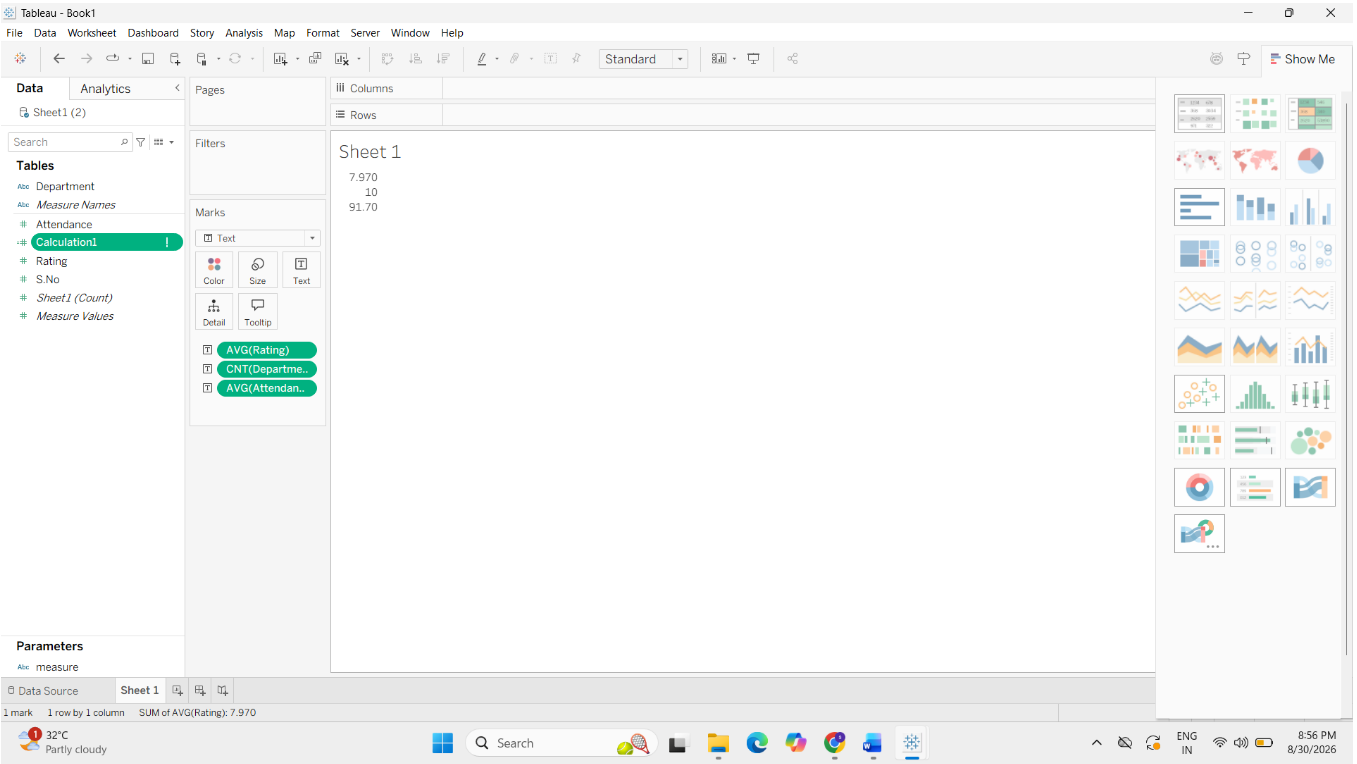
HR: Rating = **8.15**, Attendance = **93.5%**

**Interpretation**

Engineering has the highest average rating and attendance, while Support has the lowest. Parameters improve flexibility because the user can switch between Average Rating and Average Attendance in the same chart instead of creating separate static charts.

**Result**

Thus, an interactive KPI dashboard with parameter-based visualization was successfully created.




## Q3. Drill-Down Dashboard with Dashboard Actions

### Aim

To create an interactive Tableau dashboard that allows users to analyze student enrollment from Program level to Specialization level.

### Algorithm

1. Connect the student enrollment dataset to Tableau.
2. Create a Program-level enrollment bar chart.
3. Create a Specialization-level enrollment chart.
4. Add both charts to a dashboard.
5. Create a Filter Action.
6. Set the Program chart as the source.
7. Set the Specialization chart as the target.
8. Test the drill-down interaction.

### Program-wise Enrollment

CSE = **195**

ECE = **85**

MECH = **103**

**Tableau Procedure**

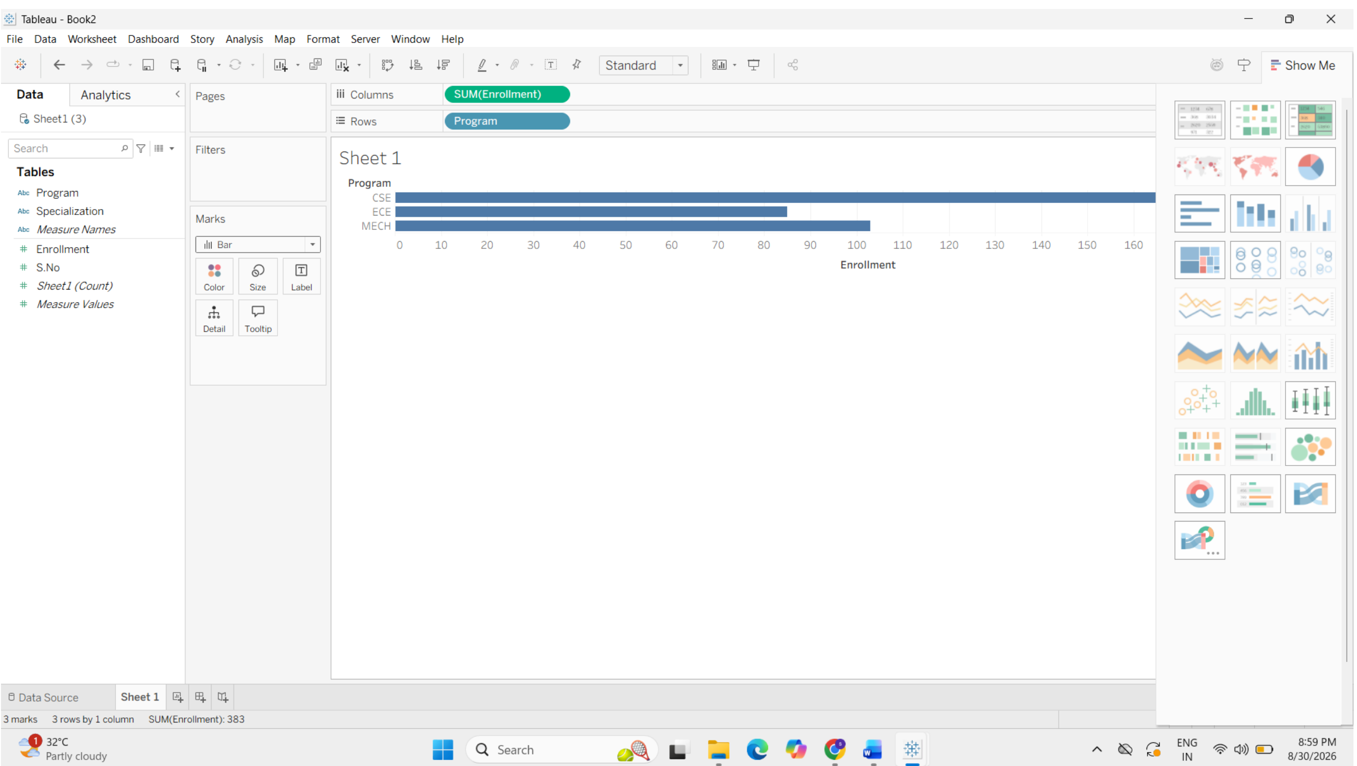
- Drag **Program** to Rows.
- Drag **Enrollment** to Columns.
- Create the Program bar chart.
- Create another chart using **Specialization** and **Enrollment**.
- Add both worksheets to the dashboard.
- Select **Dashboard → Actions → Add Action → Filter**.
- Select the Program chart as the source and Specialization chart as the target.
- Use **Program** as the filtering field.

**Analysis**

CSE has the highest enrollment with **195 students**. When the user clicks the CSE bar, only CSE specializations are displayed in the detail chart. This reduces clutter and makes exploration easier. The limitation is that users must interact with the Program chart to view specialization details, and all specializations cannot be compared simultaneously in the filtered view.

**Result**

Thus, a drill-down dashboard using dashboard filter actions was successfully created.




## Q4. Multivariate Analysis Dashboard

**Aim**

To analyze the relationship between Advertising, Website Visits and Sales using an interactive Tableau dashboard.

**Algorithm**

1. Connect the marketing dataset to Tableau.
2. Create a scatterplot of Advertising versus Sales.
3. Use Website Visits as color or size.
4. Add a trend line.
5. Create a histogram or box plot for Sales.
6. Combine the views into a dashboard.
7. Analyze the relationship and possible visual clutter.

**Tableau Procedure**

- Drag **Advertising** to Columns.
- Drag **Sales** to Rows.
- Drag **Website Visits** to Color or Size.
- Select **Analytics → Trend Line**.
- Create a second worksheet using Sales for a histogram or box plot.
- Combine both worksheets into a dashboard.

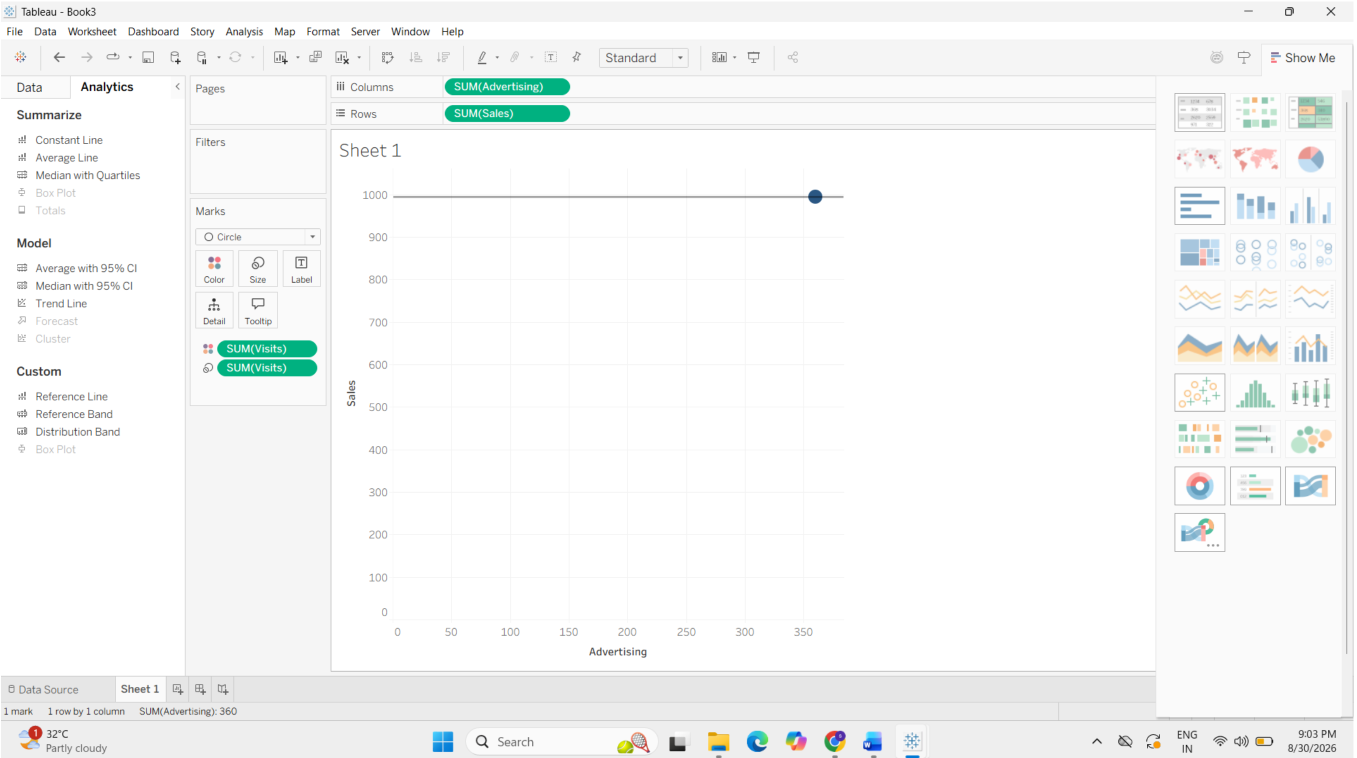
**Analysis**

The scatterplot shows a **very strong positive relationship** between Advertising and Sales. As advertising expenditure increases, sales also increase. Website Visits also increase along with advertising and sales.

The additional Website Visits encoding provides useful information about the third variable. However, using too many visual encodings can make the dashboard cluttered. Using either color or size is clearer.

**Result**

Thus, the Multivariate Analysis Dashboard was successfully created and the relationship between advertising and sales was analyzed.




## Q5. Misleading Dashboard and Redesign

### Aim

To demonstrate how inappropriate visualization choices can mislead viewers and to create a redesigned dashboard that communicates the data accurately.

### Algorithm

1. Connect the department pass percentage dataset to Tableau.
2. Create a misleading dashboard using a truncated Y-axis.
3. Use an inappropriate color scheme.
4. Give exaggerated visual emphasis to one department.
5. Create a redesigned dashboard.
6. Use an appropriate baseline.
7. Order the departments.
8. Use consistent colors and clear labels.
9. Compare both dashboards.

### Data Analysis

- CSE = **91%**

- ECE = **93%**
- EEE = **92%**
- MECH = **89%**
- CIVIL = **90%**
- AIDS = **95%**
- IT = **94%**

**Misleading Dashboard**

The misleading dashboard uses a truncated Y-axis, inappropriate colors and exaggerated emphasis on one department. These choices can make small differences appear much larger than they actually are.

**Redesigned Dashboard**

The redesigned dashboard uses:

- A suitable baseline beginning at **0%**
- Ordered categories
- Consistent color encoding
- Clear labels
- Equal visual emphasis

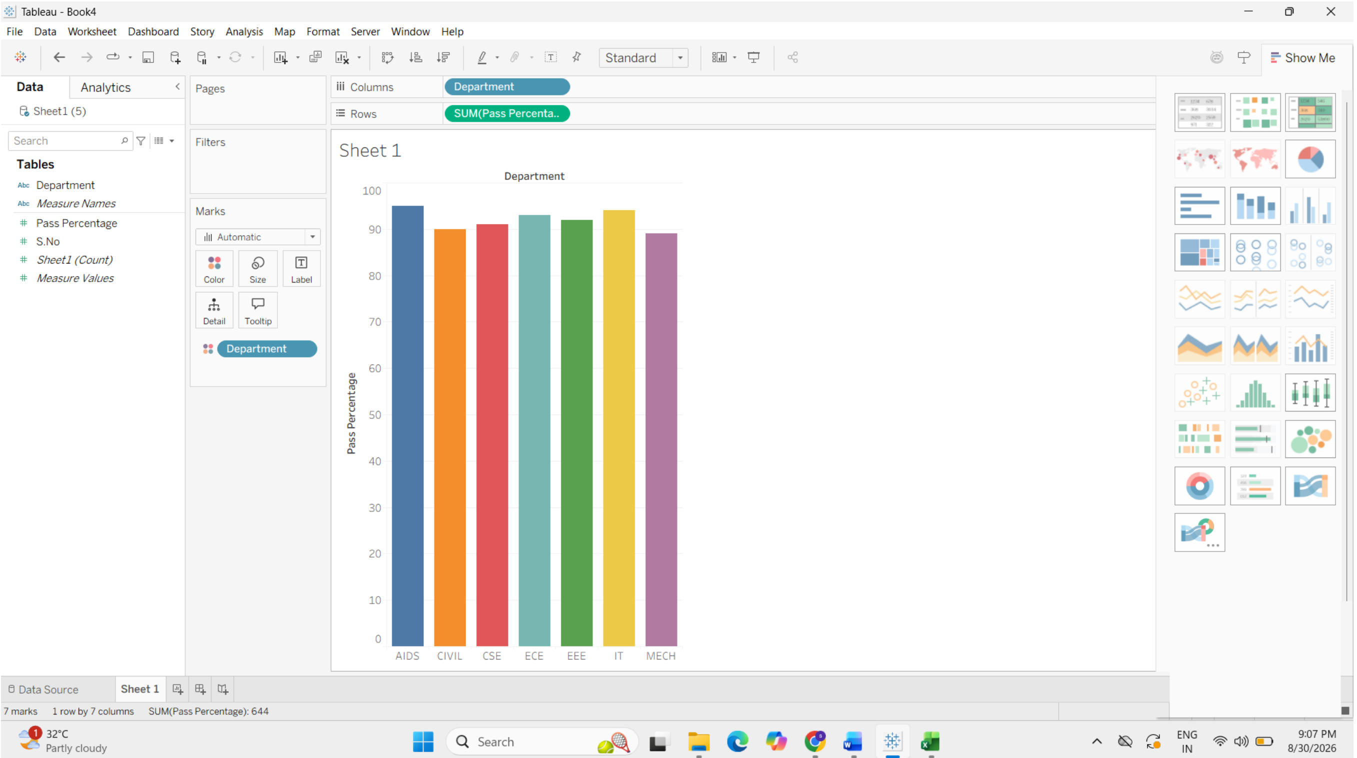
**Comparison**

The misleading dashboard can distort the viewer's perception of the data. For example, AIDS has **95%** and MECH has **89%**, which is only a **6 percentage-point difference**, but a truncated axis may make the difference appear much larger.

The redesigned dashboard presents the values honestly and allows accurate comparison between departments.

**Result**

Thus, the misleading dashboard and redesigned dashboard were successfully created, and the effect of visualization choices on data interpretation was analyzed.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Tables
Department
Measure Names
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Measure Values
Pages
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